

Guidelines

Syringe labelling in anaesthesia and critical care areas: review 2022



Association
of Anaesthetists

September 2022

Syringe labelling in anaesthesia and critical care areas: review 2022

This document replaces the Association of Anaesthetists' previous guidance¹ on this topic, following the publication of BS ISO 26825:2020.²

BS ISO 26825:2020 is the second edition of the standard for user-applied labels for syringes containing drugs used in anaesthesia. It technically revises, cancels and replaces the 2008 first edition. It gives requirements for labels attached to syringes so that the contents can be identified during anaesthesia and covers the colour, size, design and general properties of the label and the typographical characteristics of the wording for the drug name. Its purpose is solely for use in anaesthesia and as such covers a range of core drug groups. It is acknowledged that these labels may find a use in other critical care areas.

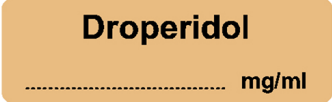
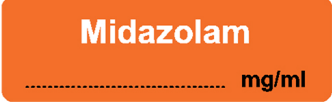
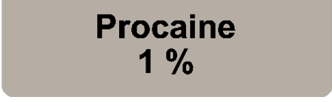
The main technical reason for the revision of BS ISO 26825 was to improve the colour, size and design of the labels. Several labels were revised to take account of comments made regarding their clarity and possibility of confusion in use. The changes are as follows:

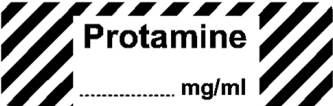
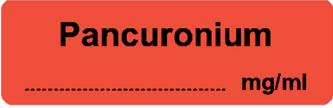
- The label design for drugs of opposite action (and antagonists) has been made more precise as it had been noted that the diagonal stripes were often minimised. As well as the original requirement for height of the diagonal stripes, it has become a requirement that the diagonal stripes each side of the drug name shall be at least 15% of the label length (Example 4).
- Muscle relaxant reversal drug labels have been made more discernible from other drug labels of opposite action (and antagonists) by the addition of a transverse black line beneath the drug name (Example 11).
- Some of the lettering colours have changed. The two drug names adrenaline and suxamethonium have been made even more prominent. Hitherto the labels for both drugs had a black background on the upper half of the label. The lettering then took the main background colour: violet in the case of adrenaline, warm red for suxamethonium. This was changed for both labels: the lettering is now white (Examples 9,14).
- In the case of benzodiazepines, a persistent comment was that the orange of the benzodiazepines had the potential to be confused with the warm red of the muscle relaxants. To counter this possibility, the lettering of the benzodiazepine labels has been changed to white (Example 3).
- The black border for the protamine label has been removed, as this seemed unnecessary, but it has been retained for heparin (Examples 7,8).
- The positioning of the drug concentration has been made clearer. With most labels this is below the drug name to the right. In the case of drugs of opposite action (and antagonists) it should not be printed over diagonal lines but below the drug name and aligned with the right-hand edge of it. In the case of pre-printed drug concentrations, it is preferred that these be below the drug name and central (Examples 4,6).
- Some drugs are premixed in the ampoule. In this case if one label is used, the background colour should reflect the intended action of the drug mixture. An example is neostigmine and glycopyrronium. The intended action is muscle relaxant reversal, and this is accomplished by neostigmine. Therefore, the label should be the same as neostigmine with glycopyrronium printed below neostigmine.
- Tall man lettering (Example 12) may be used to differentiate certain specified drugs.

¹ <https://anaesthetists.org/Home/Resources-publications/Guidelines/Archived-guidelines/Syringe-labelling-in-critical-care-areas-review>

² BS ISO 26825:2020 Anaesthetic and respiratory equipment – User-applied labels for syringes containing drugs used during anaesthesia – Colours, design and performance

Note to users: The colours are only a guide. All syringes containing drugs must be labelled. It is important to check the drug ampoule and correctly label the syringe with the correct label. Blank coloured labels are a potential source of confusion and should not be used.

Drug class	Examples	Pantone® colour	Label examples
Anti-emetics	Droperidol, metoclopramide, tropisetron	Salmon 156 Black	 Example No. 1
Anticholinergic drugs	Atropine, glycopyrronium	Green 367 Black	 Example No. 2
Benzodiazepines	Diazepam, midazolam	Orange 151 White Black	 Example No. 3
Benzodiazepine antagonists	Flumazenil	Orange 151 White Black	 Example No. 4
Induction drugs	Thiopental, methohexitone, propofol, ketamine	Process yellow C Black	 Example No. 5
Local anaesthetics	Procaine, lidocaine, bupivacaine, ropivacaine	Grey 401 Black	 Example No. 6

Miscellaneous drugs	Heparin	White Black	 Example No. 7
	Protamine	White Black	 Example No. 8
	Other drugs (e. g. amoxicillin)	Black White	No example given
Muscle relaxants	Suxamethonium	Black White Warm red	 Example No. 9
	Pancuronium, (cis)atracurium, vecuronium, rocuronium	Warm red Black	 Example No. 10
Muscle relaxant reversal drugs	Neostigmine, edrophonium, pyridostigmine, sugammadex	Warm red White Black	 Example No. 11
Opioids	Morphine, fentanyl, pethidine, sufentanil, remifentanil	Blue 297 Black	 Example No. 12
Opioid antagonists	Naloxone	Blue 297 White Black	 Example No. 13

Vasopressors	Adrenaline	Black White Violet 256	 Example No. 14
	Noradrenaline, ephedrine, phenylephrine, metaraminol	Violet 256 Black	 Example No. 15
Hypotensive drugs	Nitroprusside, nitroglycerine, phentolamine, hydralazine	Violet 256 White Black	 Example No. 16

The Pantone® Matching System is an example of a suitable product available commercially. The colours represented in the examples of labels are not to be used for colour matching.

Dr Bill Boaden

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Association of Anaesthetists*

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Association of Anaesthetists is the brand name used to refer to both the Association of Anaesthetists of Great Britain & Ireland and its related charity, AAGBI Foundation (England & Wales no. 293575 and in Scotland no. SC040697).



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